

Credit - Drishti Ias

Montreal Protocol and World Ozone Day



Source: TH

Why in News?

On the occasion of the **International Day for the Preservation of the Ozone Layer (World Ozone Day , observed on 16 th September)** , global attention was drawn to the historic journey of the **Montreal Protocol**.

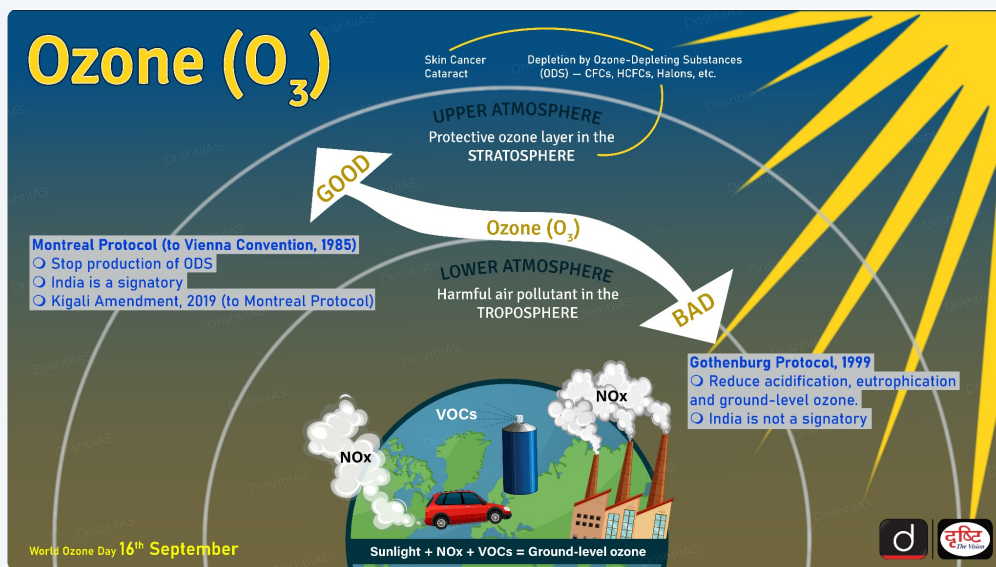
What is the Montreal Protocol?

- **Background:** In the 1970s and 1980s, scientists confirmed that certain **synthetic chemicals**, primarily **chlorofluorocarbons (CFCs)** used in refrigeration and aerosols, were migrating to the **stratosphere** (altitude range of 12 to 50 kilometers above Earth's surface) and destroying **ozone molecules**, creating a massive "**ozone hole**" over **Antarctica**.
- **Vienna Convention (1985):** Prompted by this scientific discovery, the international community adopted the **Vienna Convention for the Protection of the Ozone Layer**.
 - Established a framework for international cooperation, environmental monitoring and scientific research, paving the way for the Montreal Protocol.
 - India became a Party to the Vienna Convention for the Protection of the Ozone Layer in 1991.
- **Montreal Protocol:** Adopted in 1987, it is a **legally binding multilateral environmental agreement** aimed at protecting the ozone layer by regulating the production and consumption of **nearly 100 man-made chemicals known as Ozone Depleting Substances (ODS)**, such as chlorofluorocarbons (CFCs) and **hydrochlorofluorocarbons (HCFCs)**.
 - **India became a Party to the Montreal Protocol in June 1992.**
 - **Regulatory Metric:** Under the Protocol, "Consumption" is formally defined as **Production + Imports - Exports - Verified Destruction**.
 - **Targeted Phase-Out:** The treaty categorizes chemicals by family and sets strict, differentiated timetables for developed and developing countries to eventually eliminate their use.
 - **Success:** Often celebrated as the world's most successful environmental treaty, it has facilitated the phase-out of 99% of ODS globally.
 - India has **met its Montreal Protocol commitments** , including the **complete phase-out of HCFC-141b** from production, import and use from **1 January 2020** . It also phased out **Chlorofluorocarbons (CFCs), Carbon tetrachloride (CTC) and Halons** , achieving a **67.5% reduction in HCFC production and consumption by 2025** .

- **Universal Ratification:** On 16 th September 2009, the **Vienna Convention and the Montreal Protocol** achieved a monumental milestone by **becoming the first treaties in UN history to attain universal ratification (all 198 UN member states)**.
- **The Kigali Amendment (2016):** As industries moved away from ozone-damaging CFCs and HCFCs, they adopted **Hydrofluorocarbons (HFCs)** as alternatives.
 - While HFCs do not harm the ozone layer, they are exceptionally potent **Greenhouse Gases (GHGs)** that heavily exacerbate climate change.
 - To address this, the Parties to the Montreal Protocol adopted the **Kigali Amendment** at their 28 th Meeting of the Parties (MOP 28) in Kigali, Rwanda, in 2016.
 - It legally expanded the Protocol's mandate to **phase down non-ozone-depleting HFCs** (potent greenhouse gases with high global warming potential) turning the ozone agreement into an indispensable climate change mitigation framework by formally mandating a global phase-down of HFCs.
- **India's Commitment :** India ratified the **Kigali Amendment in 2021** , committing to **freeze HFC consumption in 2028** and phase it down by **85% by 2047** , in four stages: **10% by 2032, 20% by 2037, 30% by 2042 and 85% by 2047** against the baseline.
- **Climate Impact:** By curbing HFC emissions and pushing industries toward sustainable cooling alternatives, the successful implementation of the Kigali Amendment is projected to avoid up to **0.3-0.5°C of future global warming** by the end of this century.

What is World Ozone Day?

- **Background:** In 1994, the **United Nations General Assembly (UNGA)** officially proclaimed 16th September as the International Day for the Preservation of the Ozone Layer.
 - The date was specifically chosen to commemorate the signing of the landmark **Montreal Protocol on Substances that Deplete the Ozone Layer in 1987**.
- **Objective:** The day serves as a powerful reminder of the ozone layer's vital role in shielding life on **Earth from the Sun's harmful ultraviolet (UV) radiation** and highlights the ongoing need to prevent environmental degradation.
- **2026 Theme & Focus:**
 - **Theme:** "Global action for a cooler planet - Celebrating 10 years of sustainable cooling under the Kigali Amendment to the Montreal Protocol" .
 - **Focus:** The theme highlights rising global demand for cooling amid increasing temperatures and calls for energy-efficient, climate-friendly cooling solutions, while marking a decade of efforts to phase down potent greenhouse gases.
- **India's Net Zero Portal and NAPCC Dashboard:** Launched as digital platforms to track and showcase India's climate commitments.
 - **Net Zero Portal:** Enables industries, organisations and government bodies to voluntarily register and report progress towards net-zero emissions.
 - **NAPCC Dashboard:** Provides a centralised gateway to track progress and insights across India's National Missions on Climate Change.



Frequently Asked Questions (FAQs)

1. What is the Montreal Protocol?

The **Montreal Protocol, adopted in 1987**, is a legally binding environmental treaty that regulates and phases out **ozone-depleting substances (ODS)**.

2. What is the significance of the Vienna Convention?

The **Vienna Convention (1985)** established the framework for **international cooperation, scientific research and monitoring** for ozone protection, paving the way for the Montreal Protocol.

3. What is the Kigali Amendment?

Adopted in **2016**, it mandates the **phasedown of HFCs**, potent greenhouse gases, thereby linking **ozone protection with climate change mitigation**.

4. When did India join the Montreal Protocol and ratify the Kigali Amendment?

India became a Party to the **Montreal Protocol in 1992** and ratified the **Kigali Amendment in 2021**.

5. What is the significance of World Ozone Day?

Observed on **16 September**, it raises awareness about **ozone-layer protection, UV radiation and sustainable cooling**, commemorating the Montreal Protocol.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/6byTU3FMjpk>

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UPSC Civil Services Examination Previous Year Question (PYQ)

Q. Which one of the following is associated with the issue of control and phasing out of the use of ozone depleting substances? (2015)

- (a) Bretton Woods Conference
- (b) Montreal Protocol
- (c) Kyoto Protocol
- (d) Nagoya Protocol

Ans: (b)

Q. Consider the following statements: (2012)

Chlorofluorocarbons, known as ozone-depleting substances, are used

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurizing agents in aerosol cans

Which of the statements given above is/are correct?

- (a) 1, 2 and 3 only
- (b) 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

Ans: (c)

Q. Consider the following: (2019)

1. Carbon monoxide
2. Methane
3. Ozone
4. Sulphur dioxide

Which of the above are released into the atmosphere due to the burning of crop/biomass residue?

- (a) 1 and 2 only
- (b) 2, 3 and 4 only
- (c) 1 and 4 only
- (d) 1, 2, 3 and 4

Ans: (d)

Q. The formation of ozone holes in the Antarctic Region has been a cause of concern. What could be the reason for the formation of this hole? (2011)

- (a) Presence of prominent tropospheric turbulence; and inflow of chlorofluorocarbons
- (b) Presence of prominent polar front and stratospheric clouds; and inflow of chlorofluorocarbons
- (c) Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons
- (d) Increased temperature at polar region due to global warming

Ans: B

Pygmy Hog



Source: TH

The **Assam government recently highlighted the success of the Pygmy Hog Conservation Programme** , which has increased the pygmy hog population in the State significantly over the last three decades.

Pygmy Hog

- **About:** The **pygmy hog (*Porcula salvinia*)** is the **world's smallest wild suid** and the only surviving species of the genus **Porcula** . It is an **endemic species** of the Indian subcontinent and is closely associated with tall, wet alluvial grasslands of the **Himalayan foothills**.
- **Distribution:** Historically, it was found from **Uttar Pradesh to Assam** , including **Nepal's Terai and Bengal's Duars** . After remaining unreported for decades and being feared extinct, the **pygmy hog was rediscovered in 1971 in Assam** , with its surviving wild populations now mainly restricted to Assam, especially the Manas landscape.
- **Physical Features:** It is the smallest member of the **pig family (Suidae)**, characterised by an **extremely small body, ears and tail, short medial false hooves**, a streamlined body shape, and absence of facial warts found in many other pigs.
- **Habitat:** The pygmy hog **inhabits undisturbed tall grasslands of riverine ecosystems, comprising dense grasses, herbs, shrubs and young trees** .
 - Its fertile alluvial grassland habitat is **highly vulnerable to agricultural expansion and human settlement**, leading to habitat loss. It serves as an indicator species of healthy tall grassland ecosystems.
- **Diet:** It is an **omnivorous species** feeding on **roots, tubers, shoots, ground vegetation, worms, invertebrates and occasionally small vertebrates such as reptiles and eggs/nestlings of ground-nesting birds** .
- **Social Organisation:** Adult males are generally **solitary** , while females and juveniles live in **small family groups of about 4-6 individuals** ; males associate with females mainly during the breeding season.
- **Reproduction:** Breeding is **highly seasonal** , with births peaking during **late April-May with the onset of monsoon in Assam** . Litter size ranges from **2-6 piglets (usually 3-4)** , and both sexes construct nests throughout the year.
- **Conservation Status:**
 - **IUCN Red List : Endangered**
 - **CITES : Appendix I**

- [Wildlife \(Protection\) Act, 1972](#) : Schedule I
- **Conservation Efforts:** The **Pygmy Hog Conservation Programme (PHCP)** was initiated in **1995** , with major captive breeding efforts beginning in **1996** . The programme focuses on **captive breeding, disease prevention and reintroduction** .
- **Significance:** Conservation of pygmy hog habitat supports other threatened grassland species such as **Bengal florican, hispid hare, hog deer and greater one-horned rhinoceros** .



Read More: [Pygmy Hog](#)

Anant Nag Selected for Dadasaheb Phalke Award 2024



Source: PIB

Veteran Kannada actor **Anant Nag** has been selected for the prestigious **Dadasaheb Phalke Award** , India's highest honour in cinema, making him the **second Kannada actor** to receive the award after **Dr. Rajkumar (1995)** .

- He will be presented with the award at the **72nd National Film Awards Ceremony** on **22nd September 2026 at Kevadia, Gujarat** .

Anant Nag

- **Birth:** Born as **Anant Nagarkatte in 1948 in Karnataka** , he has acted in **300+ films**, including **over**

250 Kannada films , across mainstream and parallel cinema.

- **Major Works:** He made his Kannada film debut with **Sankalpa** and Hindi debut with **Ankur** . His notable works include **Malgudi Days** and several acclaimed Kannada films.
- **Recognition:** He has received **five Filmfare Awards** , **five Karnataka State Film Awards** , Karnataka Rajyotsava Award and Karnataka Sahitya Academy Award. He was honoured with the **Padma Bhushan** in 2025 .
- **Contributions:** He also served as a **Member of Karnataka Legislative Council (1988-1994)** and **Member of Legislative Assembly (1994-1999)** . He also served as **Minister of State for Bangalore City Development** .



Dadasaheb Phalke Award

- Instituted in **1969** , the award is India's **highest honour in the field of cinema** .
- It was introduced to commemorate **Dadasaheb Phalke** , the director of India's first full-length feature film **Raja Harishchandra (1913)** .
- The award recognises **outstanding contribution to the growth and development of Indian cinema** and includes a **Swarna Kamal (Golden Lotus) medallion, shawl and ₹10 lakh cash prize** .

Read More: [Dadasaheb Phalke Award](#)